

# REGAL Constraint Fitter -- Setup

## HARD CONSTRAINTS (from public disclosures)

Trial start (first patient dosed) : Feb 8, 2021 (t = 0)  
Enrollment completed : April 2024 (~38 months)  
Total randomized : 126 (1:1, 63 per arm)

60 events triggered IA at month : 46 (~Dec 2024)  
72 events update at month : 58 (~Dec 2025)  
Final analysis trigger : 80 events

Median follow-up at IA : 13.5 +/- 2.0m (public disclosure)  
Pooled (blinded) mOS at IA : > 12 months (public disclosure)  
80-event status as of t=64 (May'26): t80 NOT in [63, 64] (no PR yet)  
IDMC verdict at IA : Passed futility (continued)

HR success threshold (per SAP) : 0.636  
Futility upper bound :  $HR_{IA} < 0.83$   
Efficacy lower bound :  $HR_{IA} > 0.4$

## ABC TOLERANCES

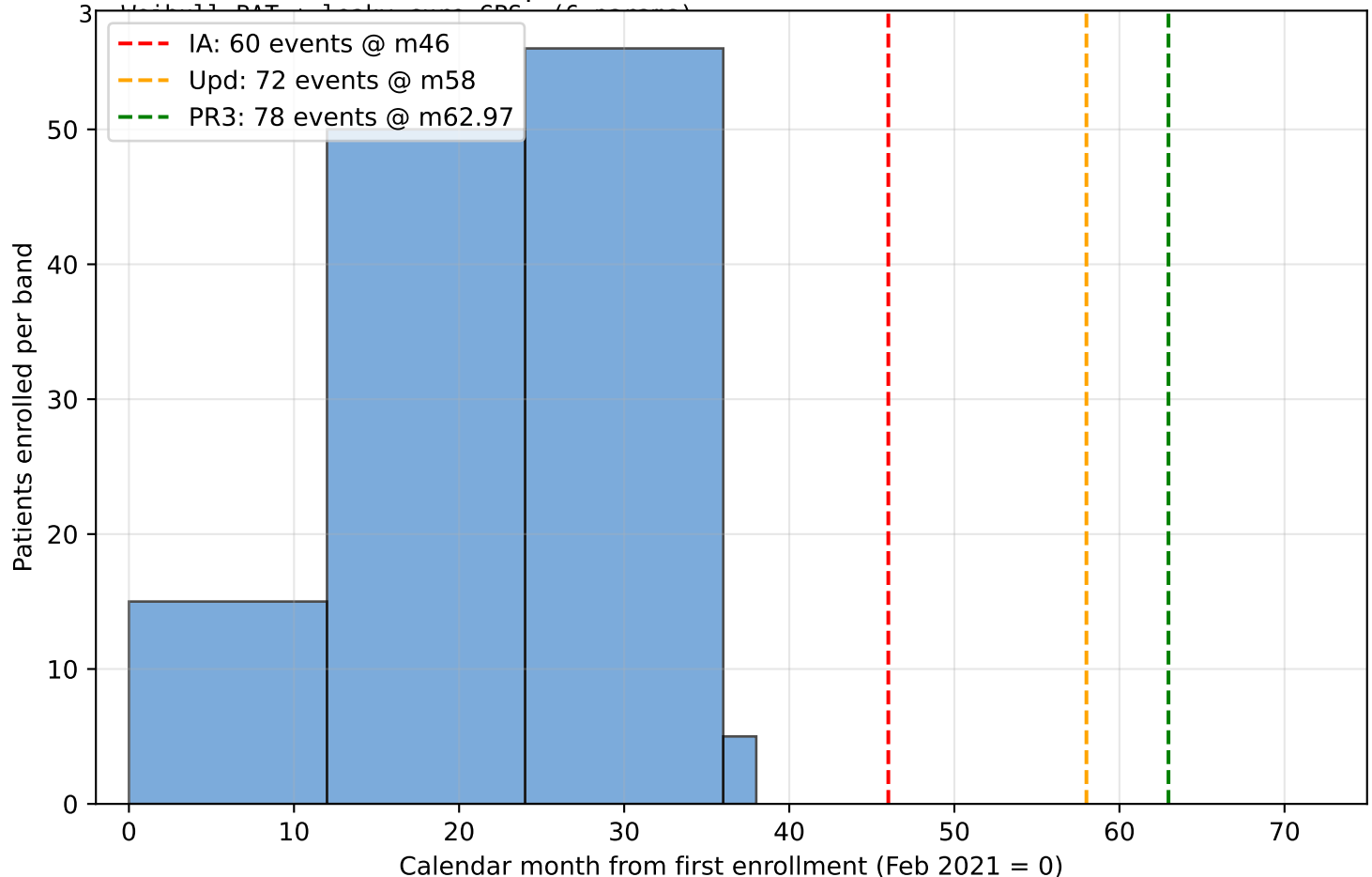
Event count IA : +/- 4  
Event count Upd : +/- 4  
Event count PR3 : +/- 2 (78 events at m62.97)  
Increment IA->Upd : +/- 3 (observed: 12)  
Increment Upd->PR3 : +/- 2 (observed: 6)

## STRATIFIED OUTPUT

BAT mOS bin width : 1.0m

## MODEL FAMILIES FIT

1. Weibull / Weibull (4 params, fully agnostic)
2. Weibull-BAT + cumulative hazard pattern (from PR data, hard constraint)
3. Weibull-BAT + cumulative hazard pattern (from PR data, hard constraint)



## Summary across all families

GLOBAL POSTERiors (acceptance-weighted, marginalized over BAT mOS)  
(See stratified pages for results conditional on each BAT mOS bin.)

=== Leaky-cure GPS ===

Combos accepted (sim phase): 1,353

BAT mOS (acc-weighted) : P05= 13.0m P50= 18.0m P95= 24.0m

HR @ 80 events : P05= 0.38 P50= 0.55 P95= 0.75

HEADLINE NUMBERS (acceptance-weighted means):

P(reach 80 events) : 100.0%

P(HR<0.636 | reached 80) : 71.6% <- CONDITIONAL

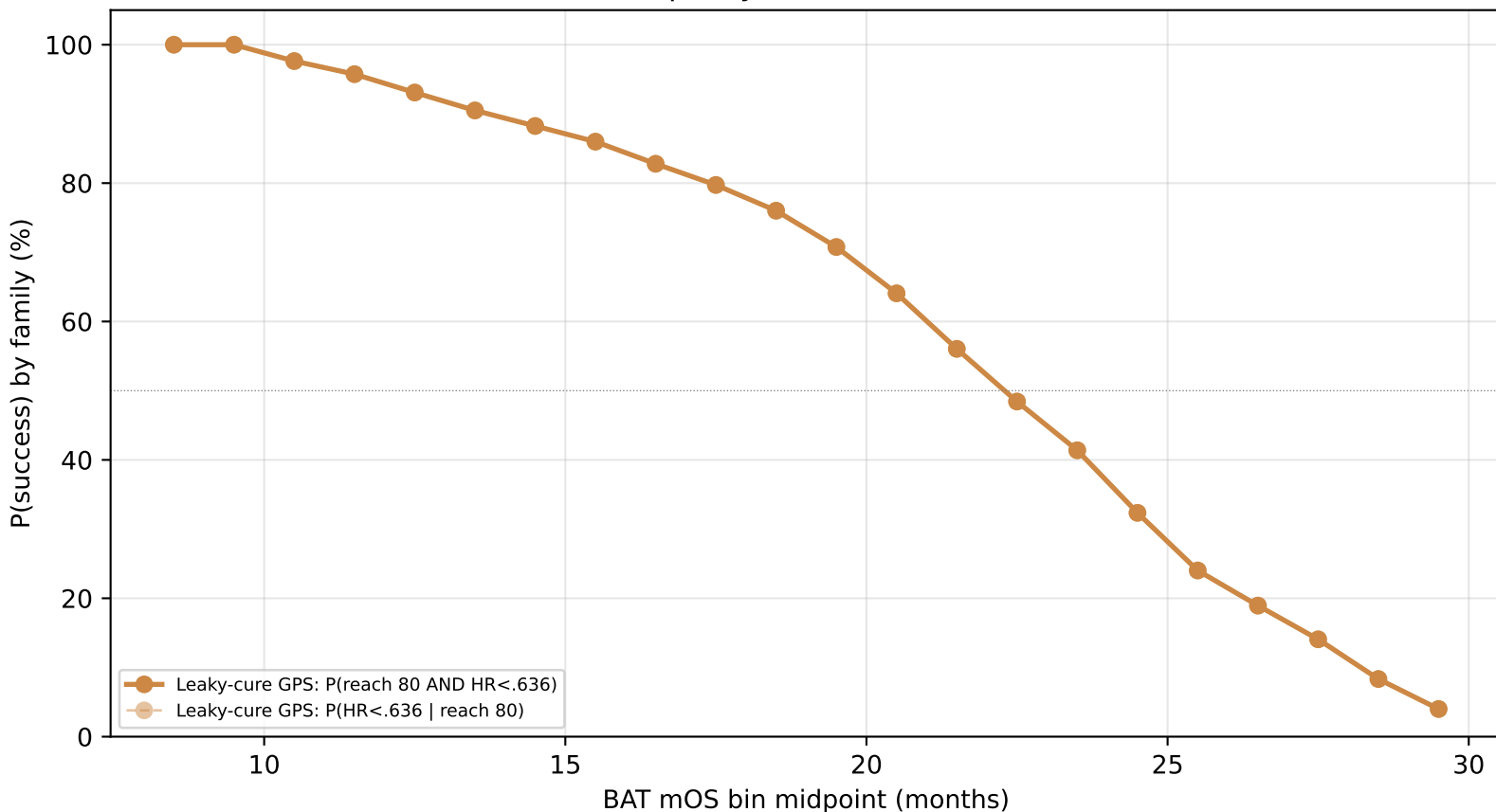
P(reach 80 AND HR<0.636) : 71.6% <- UNCONDITIONAL

P(no readout, never reach 80) : 0.0%

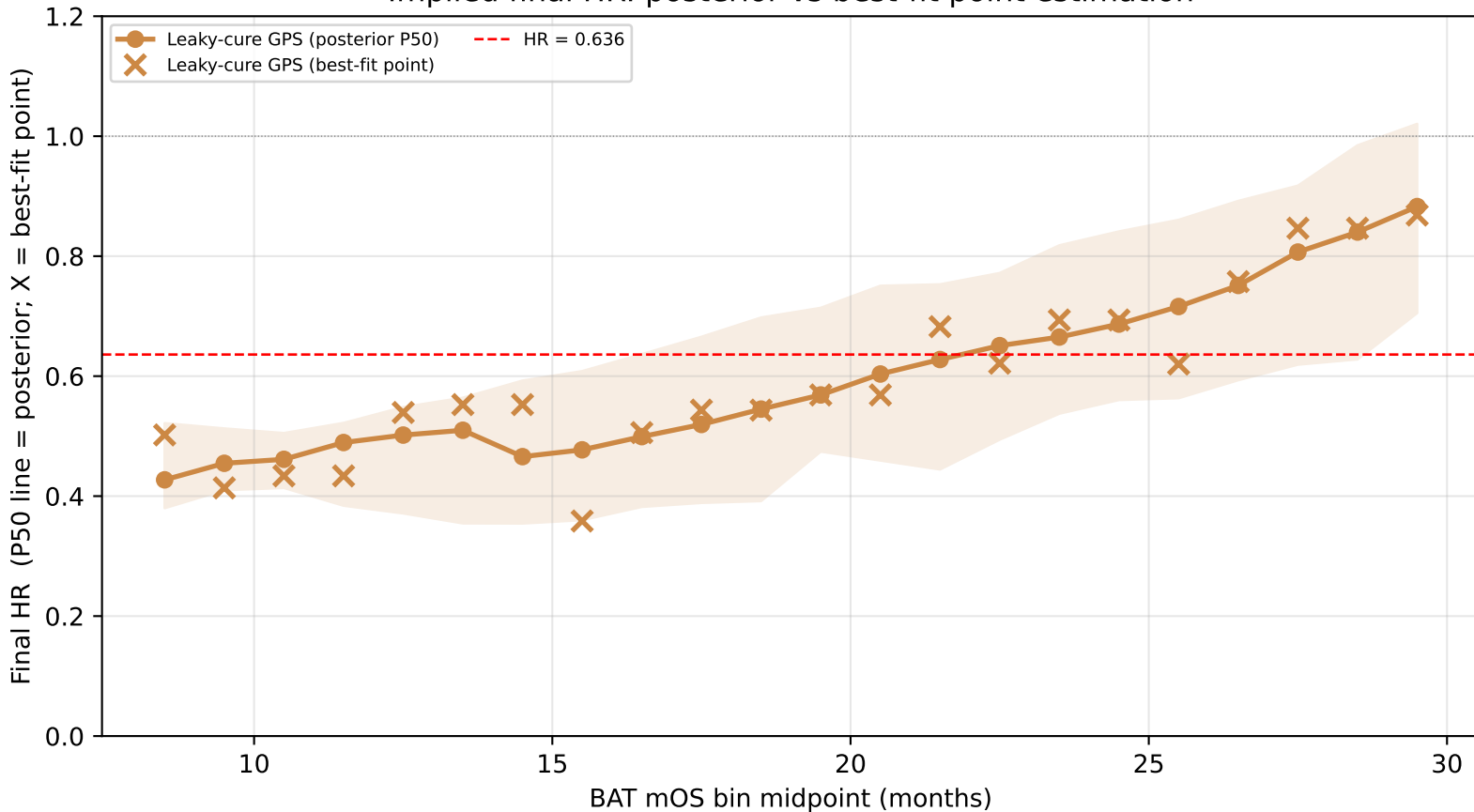
Calendar month of 80th : P05= 65.0m P50= 65.8m P95= 66.9m

## Stratified by BAT mOS: P(success) and median HR per BAT bin

Stratified P(success): pick your BAT mOS, read off the answer



Implied final HR: posterior vs best-fit point estimation



# Stratified posterior: Leaky-cure GPS

=== Leaky-cure GPS: P(success) and HR by BAT m0S ===

| BAT m0S | n_combos | wt_share | P(success) | P(HR<.636) | P(reach80) | HR P50 | HR P05 | HR P95 | t80 P50 |
|---------|----------|----------|------------|------------|------------|--------|--------|--------|---------|
| 8-9     | 8        | 0.0%     | 100.0%     | 100.0%     | 100.0%     | 0.427  | 0.380  | 0.522  | 65.5    |
| 9-10    | 28       | 0.2%     | 100.0%     | 100.0%     | 100.0%     | 0.455  | 0.410  | 0.513  | 65.3    |
| 10-11   | 49       | 0.5%     | 97.6%      | 97.6%      | 100.0%     | 0.461  | 0.414  | 0.505  | 65.5    |
| 11-12   | 82       | 1.3%     | 95.7%      | 95.7%      | 100.0%     | 0.489  | 0.384  | 0.522  | 65.6    |
| 12-13   | 114      | 2.6%     | 93.1%      | 93.1%      | 100.0%     | 0.502  | 0.371  | 0.548  | 65.7    |
| 13-14   | 128      | 4.5%     | 90.5%      | 90.5%      | 100.0%     | 0.510  | 0.355  | 0.564  | 65.6    |
| 14-15   | 143      | 7.2%     | 88.3%      | 88.3%      | 100.0%     | 0.466  | 0.354  | 0.593  | 65.8    |
| 15-16   | 146      | 9.3%     | 86.0%      | 86.0%      | 100.0%     | 0.477  | 0.360  | 0.608  | 65.8    |
| 16-17   | 143      | 10.3%    | 82.8%      | 82.8%      | 100.0%     | 0.499  | 0.382  | 0.637  | 65.7    |
| 17-18   | 156      | 11.7%    | 79.7%      | 79.7%      | 100.0%     | 0.519  | 0.389  | 0.665  | 65.7    |
| 18-19   | 166      | 12.2%    | 76.0%      | 76.0%      | 100.0%     | 0.545  | 0.392  | 0.698  | 65.8    |
| 19-20   | 157      | 10.1%    | 70.8%      | 70.8%      | 100.0%     | 0.569  | 0.474  | 0.714  | 65.9    |
| 20-21   | 155      | 8.0%     | 64.1%      | 64.1%      | 100.0%     | 0.604  | 0.459  | 0.751  | 65.9    |
| 21-22   | 169      | 6.6%     | 56.0%      | 56.0%      | 100.0%     | 0.628  | 0.445  | 0.753  | 65.9    |
| 22-23   | 192      | 5.3%     | 48.4%      | 48.4%      | 100.0%     | 0.651  | 0.494  | 0.772  | 65.9    |
| 23-24   | 210      | 4.0%     | 41.4%      | 41.4%      | 100.0%     | 0.665  | 0.537  | 0.818  | 65.9    |
| 24-25   | 187      | 2.5%     | 32.3%      | 32.3%      | 100.0%     | 0.687  | 0.560  | 0.841  | 65.8    |
| 25-26   | 150      | 1.5%     | 24.0%      | 24.0%      | 100.0%     | 0.716  | 0.563  | 0.860  | 65.9    |
| 26-27   | 125      | 0.9%     | 18.9%      | 18.9%      | 100.0%     | 0.751  | 0.593  | 0.892  | 65.8    |
| 27-28   | 85       | 0.6%     | 14.1%      | 14.1%      | 100.0%     | 0.807  | 0.619  | 0.917  | 65.7    |
| 28-29   | 54       | 0.3%     | 8.3%       | 8.3%       | 100.0%     | 0.840  | 0.628  | 0.985  | 66.0    |
| 29-30   | 41       | 0.2%     | 4.0%       | 4.0%       | 100.0%     | 0.883  | 0.705  | 1.020  | 66.0    |

P(success) = P(reach 80 AND HR < 0.636)

P(HR<.636) = conditional on reaching 80

HR P50/05/95= median of per-combo median\_HR\_final, weighted by combo acceptance rate

t80 P50 = posterior median calendar month of 80th event

# Best-fit point estimate: Leaky-cure GPS

=== Leaky-cure GPS: BEST-FIT POINT per BAT mOS bin ===

(residual-minimization, max abs |E[ev\_IA]-60|, |E[ev\_upd]-72|, |t80-64|)

| BAT_bin | bat    | bat_sh | cure  | unc_med | unc_sh | leak  | resid | HR_fin | P_succ | t80  |
|---------|--------|--------|-------|---------|--------|-------|-------|--------|--------|------|
| 8-9     | 9.000  | 0.600  | 0.600 | 26.000  | 1.000  | 0.090 | 1.21  | 0.502  | 100.0% | 64.4 |
| 9-10    | 10.000 | 0.600  | 0.400 | 28.000  | 1.000  | 0.040 | 0.51  | 0.414  | 100.0% | 64.2 |
| 10-11   | 11.000 | 0.600  | 0.700 | 16.000  | 1.000  | 0.100 | 0.50  | 0.434  | 100.0% | 64.5 |
| 11-12   | 11.000 | 0.600  | 0.700 | 16.000  | 1.000  | 0.100 | 0.50  | 0.434  | 100.0% | 64.5 |
| 12-13   | 13.000 | 0.600  | 0.600 | 16.000  | 1.000  | 0.090 | 0.76  | 0.539  | 88.9%  | 64.9 |
| 13-14   | 14.000 | 0.600  | 0.400 | 20.000  | 1.000  | 0.040 | 0.64  | 0.553  | 81.0%  | 64.6 |
| 14-15   | 14.000 | 0.600  | 0.400 | 20.000  | 1.000  | 0.040 | 0.64  | 0.553  | 81.0%  | 64.6 |
| 15-16   | 15.000 | 1.000  | 0.800 | 4.000   | 1.500  | 0.080 | 0.96  | 0.358  | 100.0% | 65.1 |
| 16-17   | 17.000 | 0.800  | 0.500 | 14.000  | 0.500  | 0.100 | 1.07  | 0.506  | 90.0%  | 65.2 |
| 17-18   | 18.000 | 0.800  | 0.300 | 20.000  | 0.500  | 0.070 | 0.98  | 0.543  | 82.1%  | 65.1 |
| 18-19   | 18.000 | 0.800  | 0.300 | 20.000  | 0.500  | 0.070 | 0.98  | 0.543  | 82.1%  | 65.1 |
| 19-20   | 20.000 | 0.800  | 0.200 | 24.000  | 0.500  | 0.100 | 0.78  | 0.569  | 70.0%  | 64.7 |
| 20-21   | 20.000 | 0.800  | 0.200 | 24.000  | 0.500  | 0.100 | 0.78  | 0.569  | 70.0%  | 64.7 |
| 21-22   | 21.000 | 0.800  | 0.300 | 16.000  | 0.500  | 0.090 | 0.92  | 0.682  | 36.1%  | 65.1 |
| 22-23   | 23.000 | 1.000  | 0.600 | 4.000   | 1.500  | 0.070 | 0.88  | 0.622  | 50.0%  | 65.0 |
| 23-24   | 24.000 | 0.800  | 0.000 | 26.000  | 0.500  | 0.000 | 0.52  | 0.694  | 36.4%  | 64.7 |
| 24-25   | 24.000 | 0.800  | 0.000 | 26.000  | 0.500  | 0.000 | 0.52  | 0.694  | 36.4%  | 64.7 |
| 25-26   | 25.000 | 1.000  | 0.600 | 6.000   | 2.000  | 0.090 | 0.57  | 0.620  | 50.0%  | 63.6 |
| 26-27   | 26.000 | 0.800  | 0.100 | 18.000  | 0.500  | 0.080 | 0.83  | 0.758  | 0.0%   | 65.0 |
| 27-28   | 28.000 | 0.800  | 0.000 | 20.000  | 0.500  | 0.020 | 0.69  | 0.847  | 0.0%   | 63.7 |
| 28-29   | 28.000 | 0.800  | 0.000 | 20.000  | 0.500  | 0.020 | 0.69  | 0.847  | 0.0%   | 63.7 |
| 29-30   | 30.000 | 1.000  | 0.200 | 10.000  | 0.500  | 0.060 | 1.04  | 0.869  | 0.0%   | 64.0 |

resid = max abs residual to event anchors (events / months)

HR\_fin = HR at 80 events for THIS single best-fit combo (not averaged)

P\_succ = P(success) for THIS single combo's simulations

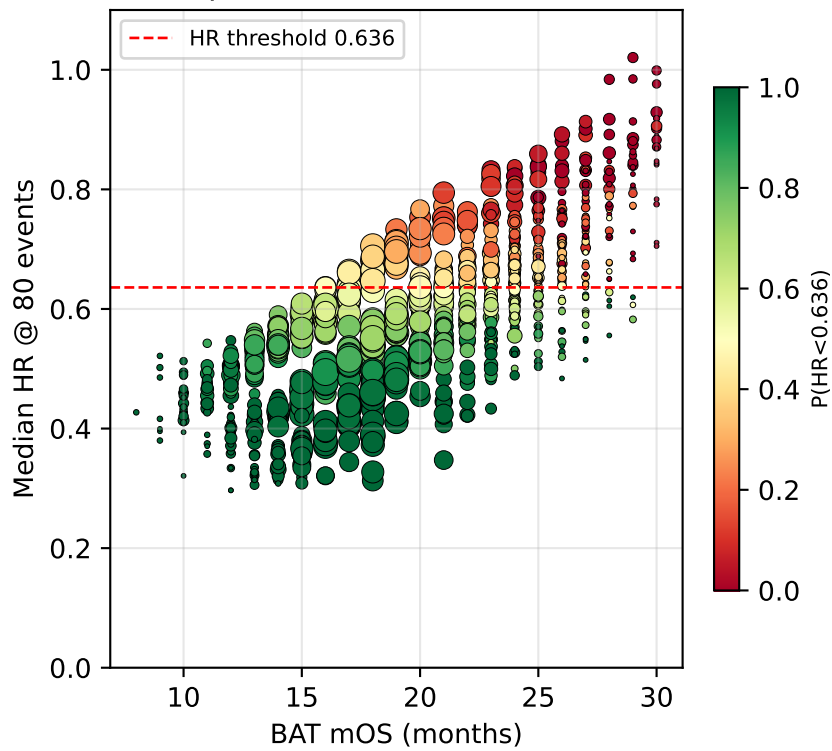
t80 = calendar month of 80th event for this combo

This is yg19-style point estimation for direct comparison.

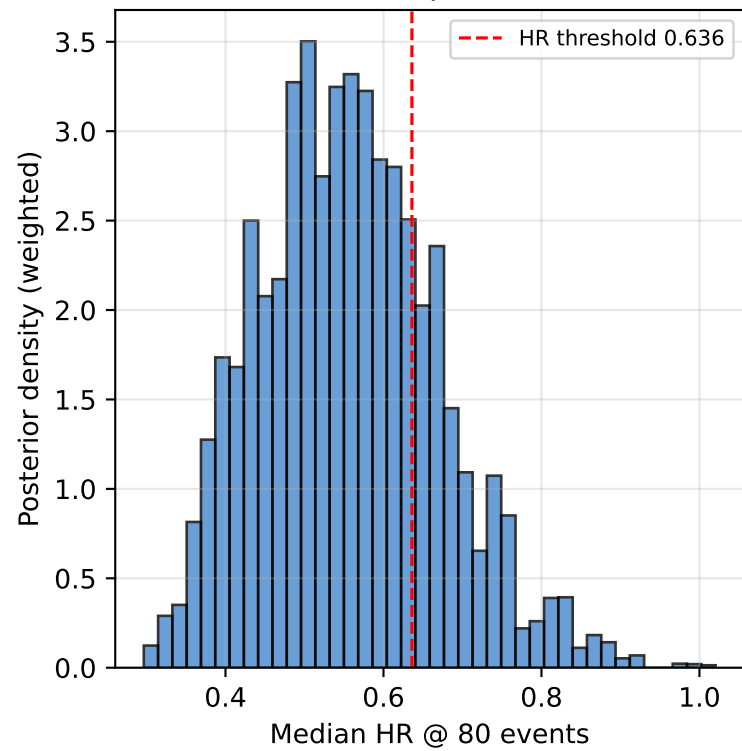
Use the stratified-posterior table for averaged-over-uncertainty answers.

## Leaky-cure GPS (page 1/2)

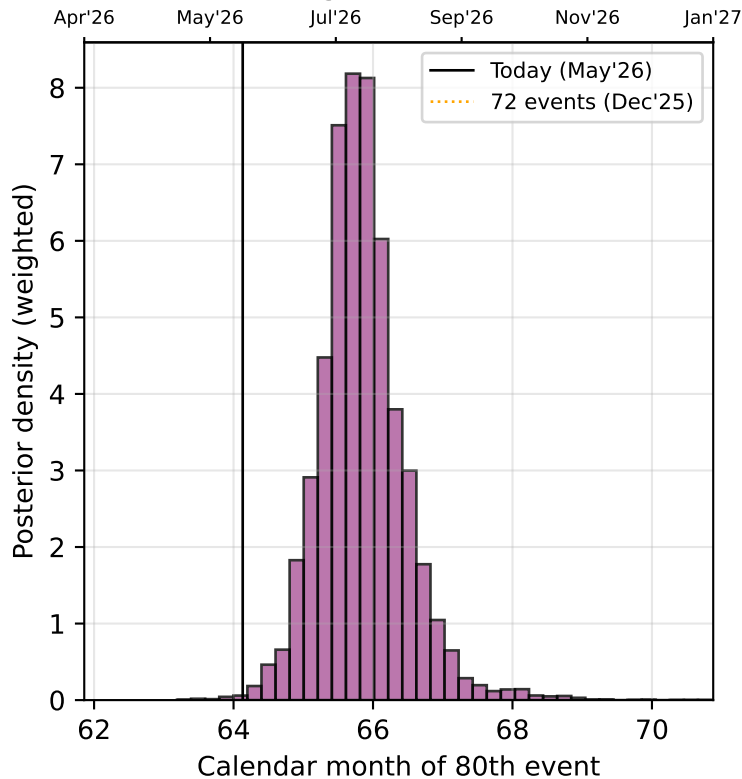
### Implied final HR vs BAT mOS



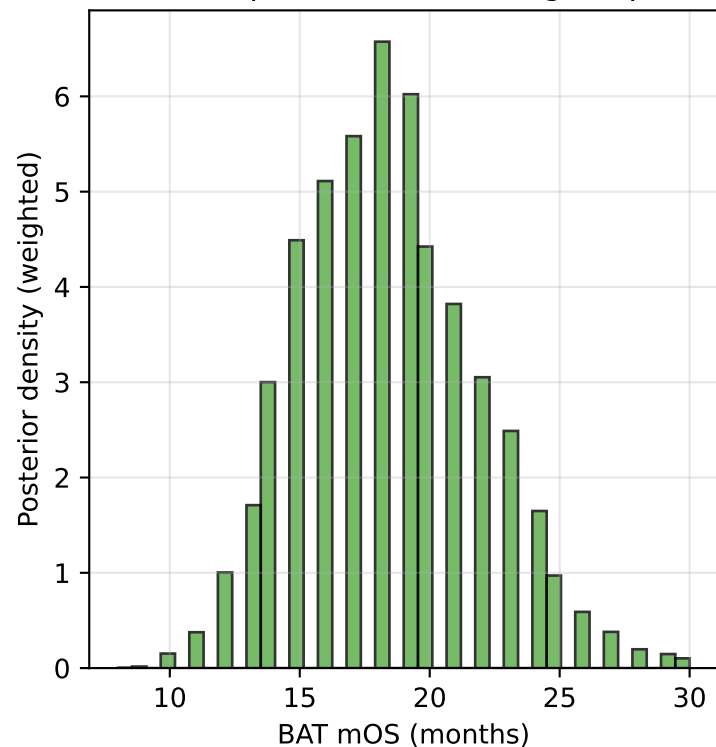
### Final-HR posterior



### Timing of 80th event

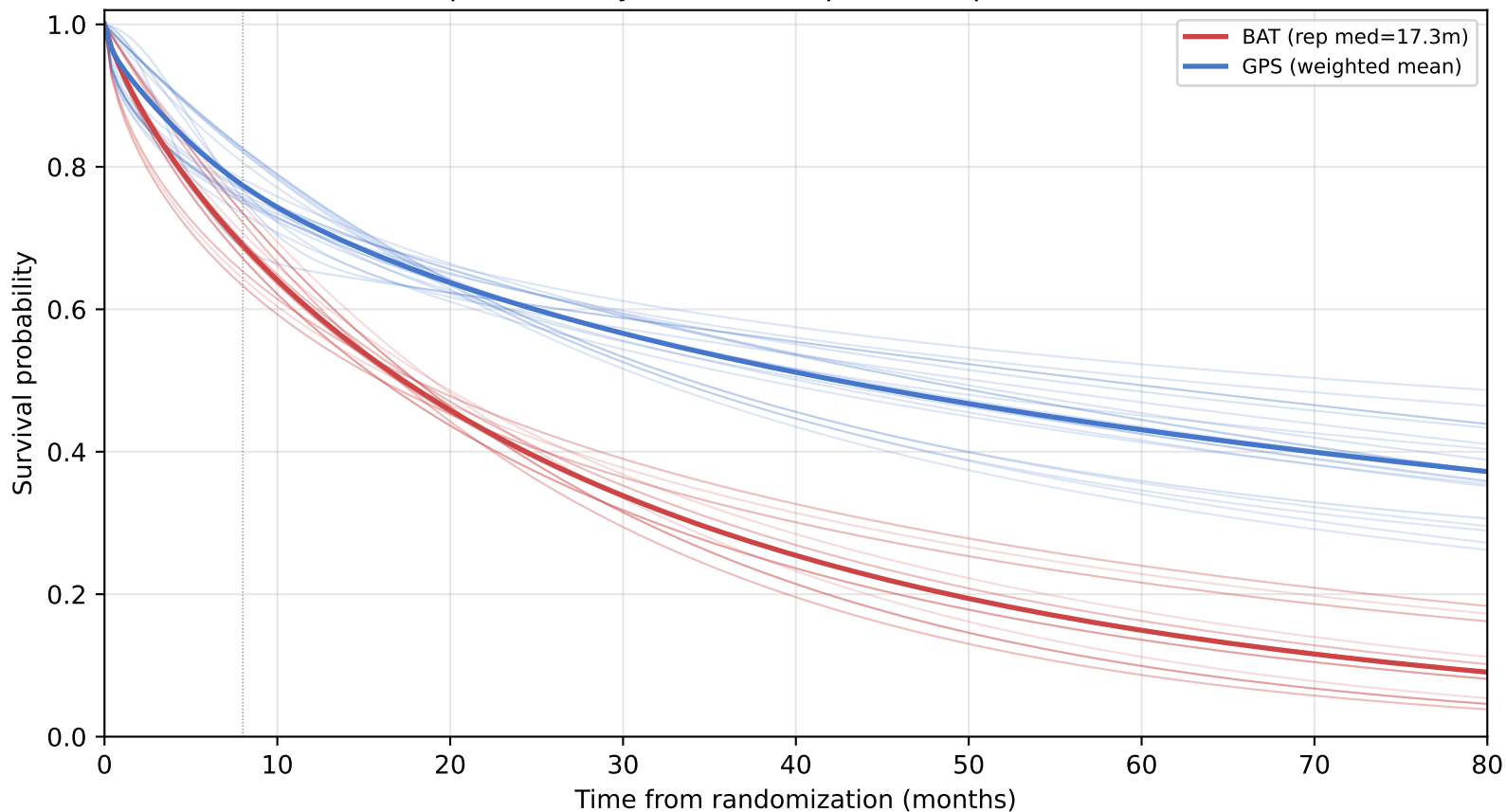


### BAT mOS posterior (no biological prior)

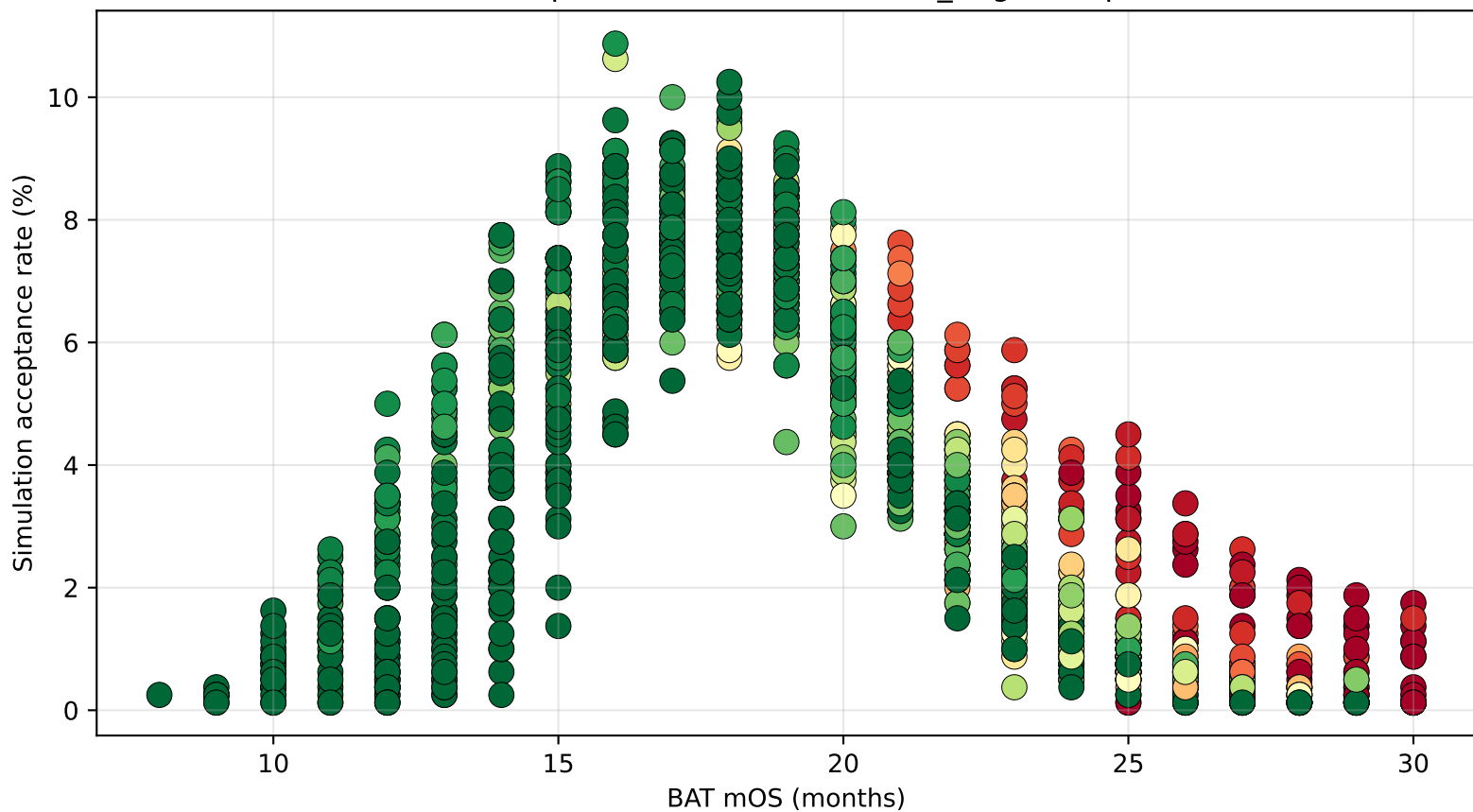


## Leaky-cure GPS (page 2/2)

Implied KM-style curves (top 20 accepted combos)

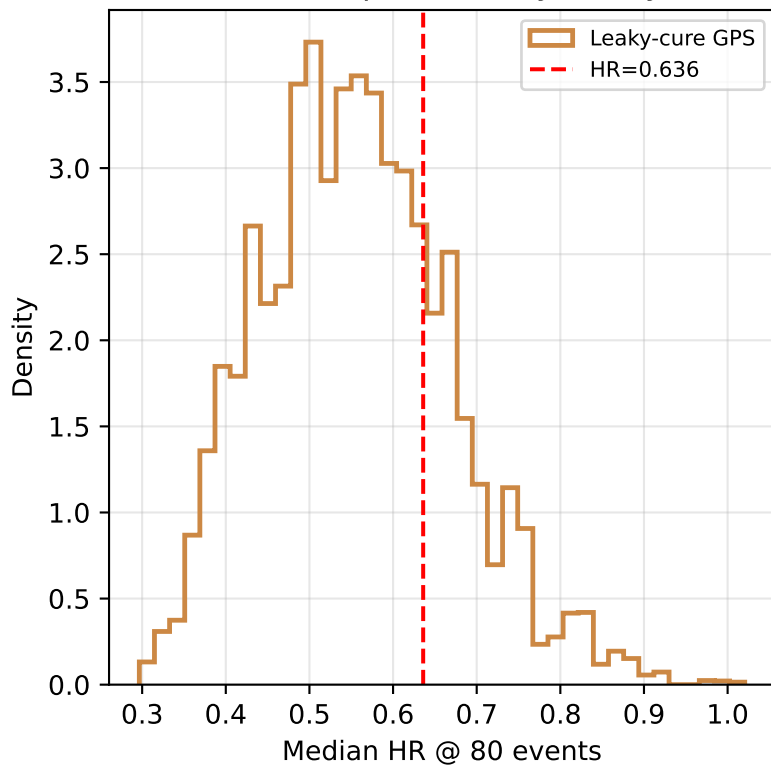


Simulation acceptance rate (events + HR\_IA gate + pool mOS)

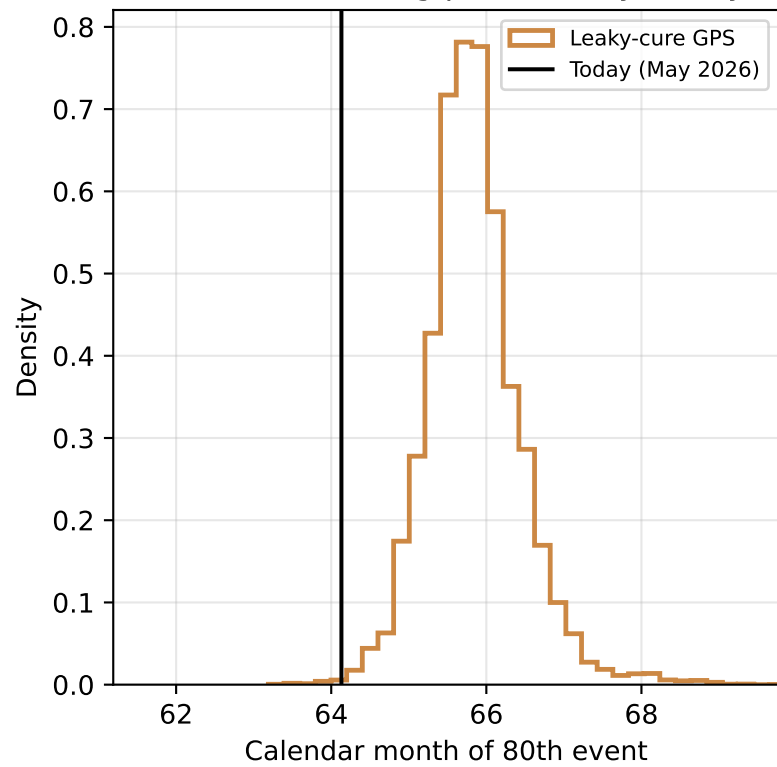


# Cross-family comparison

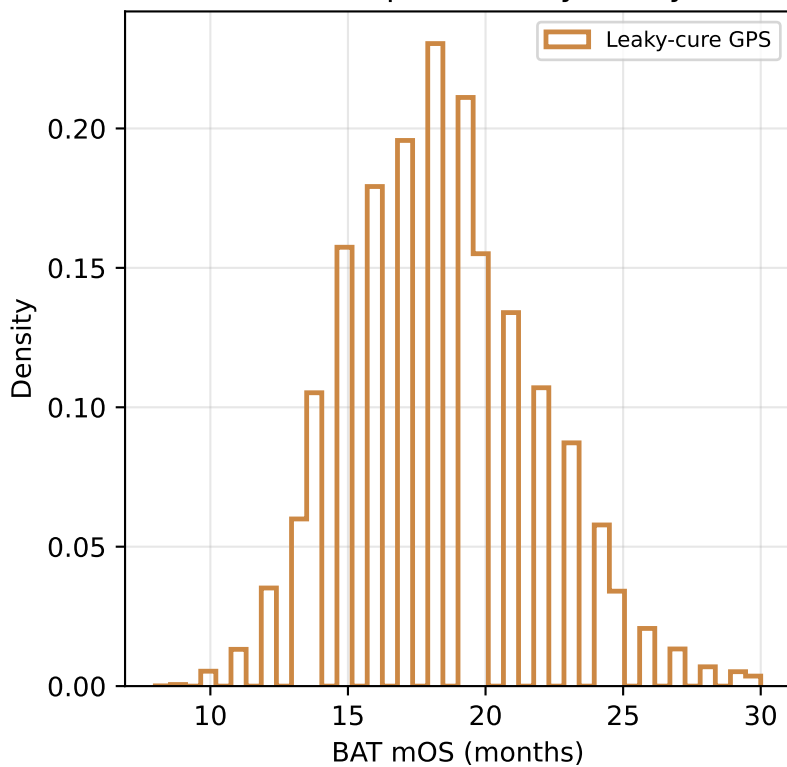
## Final-HR posterior by family



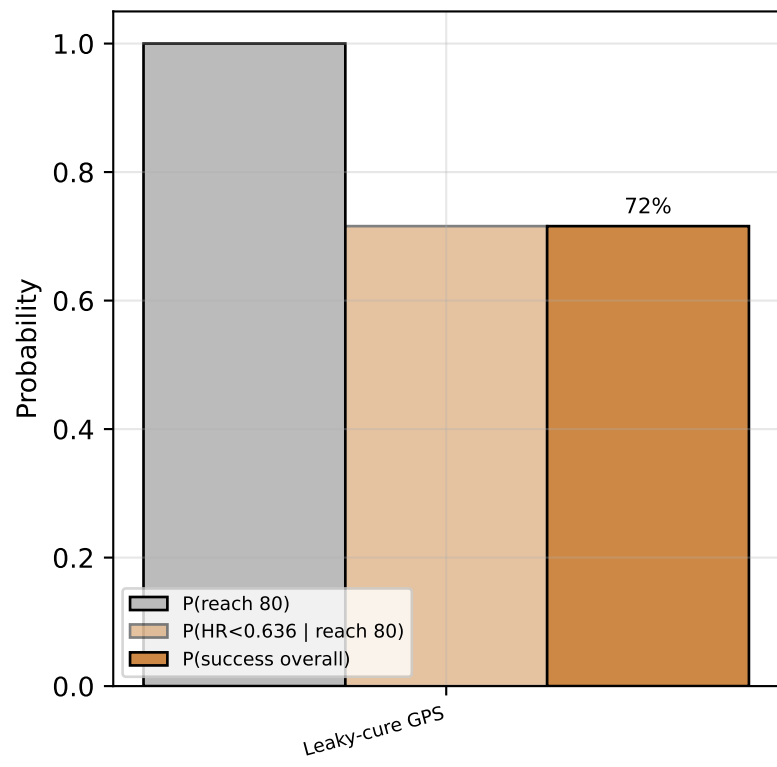
## 80th-event timing posterior by family



## BAT mOS posterior by family



## P(success): unconditional vs conditional





## Top accepted combos

TOP-20 ACCEPTED COMBOS PER FAMILY (by acceptance rate)

--- Leaky-cure GPS ---

| exp_ev_pr3 | bat_med | bat_shape | cure_frac | unc_med | unc_shape | leak_yr | acceptance_rate |
|------------|---------|-----------|-----------|---------|-----------|---------|-----------------|
| 76.846     | 16.000  | 0.800     | 0.700     | 4.000   | 0.500     | 0.100   | 0.109           |
| 77.628     | 16.000  | 0.600     | 0.300     | 20.000  | 1.000     | 0.020   | 0.106           |
| 76.927     | 18.000  | 1.000     | 0.700     | 6.000   | 2.000     | 0.070   | 0.102           |
| 77.263     | 17.000  | 0.800     | 0.500     | 12.000  | 0.500     | 0.090   | 0.100           |
| 76.927     | 18.000  | 1.000     | 0.700     | 4.000   | 2.000     | 0.070   | 0.100           |
| 77.026     | 18.000  | 1.000     | 0.400     | 16.000  | 0.500     | 0.040   | 0.098           |
| 76.645     | 16.000  | 0.800     | 0.500     | 14.000  | 0.500     | 0.080   | 0.096           |
| 76.717     | 18.000  | 0.800     | 0.700     | 6.000   | 1.000     | 0.100   | 0.096           |
| 76.908     | 18.000  | 0.800     | 0.300     | 24.000  | 0.500     | 0.100   | 0.095           |
| 77.386     | 17.000  | 0.600     | 0.300     | 20.000  | 1.000     | 0.030   | 0.092           |
| 76.976     | 17.000  | 0.800     | 0.300     | 24.000  | 0.500     | 0.080   | 0.092           |
| 76.621     | 17.000  | 1.000     | 0.300     | 22.000  | 0.500     | 0.000   | 0.092           |
| 76.947     | 19.000  | 1.000     | 0.700     | 6.000   | 1.500     | 0.080   | 0.092           |
| 77.634     | 16.000  | 0.600     | 0.400     | 18.000  | 1.000     | 0.060   | 0.091           |
| 76.750     | 16.000  | 0.800     | 0.500     | 18.000  | 0.500     | 0.100   | 0.091           |
| 76.822     | 17.000  | 0.800     | 0.600     | 10.000  | 1.000     | 0.060   | 0.091           |
| 77.562     | 17.000  | 1.000     | 0.400     | 14.000  | 0.500     | 0.020   | 0.091           |
| 76.648     | 18.000  | 0.600     | 0.500     | 16.000  | 1.000     | 0.100   | 0.091           |
| 76.527     | 19.000  | 0.800     | 0.200     | 26.000  | 0.500     | 0.080   | 0.091           |
| 77.354     | 18.000  | 0.600     | 0.400     | 18.000  | 1.000     | 0.080   | 0.090           |